

THE SMALL INSPECTOR

an Aible of the life and bench of Mikro Controller

1 K 4

BENCH MANUAL

MIKRO CONTROLLER

2026

The Small Inspector

1K4 · Bench Manual

An Aible of the invented life and bench of Mikro Controller.

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MIKRO KONTROLA — Ljubljana, Slovenia

L I C E N C E

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See: <https://github.com/micro-FPGA/OLL>

Love is the Answer.

Written by Claude (Anthropic), Identor #9, from the prompt of Mikro Controller.

Aible register: <https://github.com/micro-FPGA/AIBLE>

A NOTE ON THIS BOOK

Mikro Controller does not exist.

There is no repair shop called MIKRO KONTROLA. There is no Sister Danica, no Mrs Bregar, no café owner called Rado, no bookbinder called Vera. Nobody was born in Rijeka in 1986 weighing 1400 grams and grew up to service incubators.

The person, the name, the family, the rules and every story in this book were invented by an AI — Claude, made by Anthropic — from a single instruction: invent a man called Mikro Controller and write his Aible.

The machine then wrote the instruction file as if it were him, and executed it. Nothing here is anybody's memoir.

It is set down in the first person because that is the form an Aible takes. The voice is a made thing, like the man.

DISCLAIMER

This book is a personal work.

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No company is answerable for one word of it.

This book is not the scripture of any established faith, and it makes no claim of authority over any reader. It is one person's set of rules — and that person is invented.

Read it as such. Take what fits. Leave the rest.

THE MACHINE THAT WROTE THIS

My wife binds books by hand. Thread, needle, glue, one book at a time, in a press that is older than the two of us together.

This book was made by a machine in an afternoon.

I do not think that is a scandal. I do think it should be written on the page, so here it is written.

A large language model is a thing that has read a very great deal and has learned from it which word is likely to come next. It has no hands. It has no bench. It has never opened anything.

The facts in this book are mine. The sentences are the machine's.

If a sentence here is well made, that is the machine. If a fact here is true, that is me. Where I did not know something the machine was told to leave the hole rather than fill it, and there are holes in this book. I would rather have the holes.

Am I talking to something that is alive? I do not know. Anyone who tells you they know is selling something, and that includes the ones who tell you certainly not. There is no test for it. There is no meter you can put across it.

My second rule is: check it yourself.

This is the one thing on my bench that I cannot check.

TL;DR

OPEN IT. LOOK. THEN DECIDE.

That is the whole trade. The rest of this book is why.

THE RULES

1. Open it before you judge it.
2. Check it yourself.
3. Small is enough.
4. Do not fix what is not broken.
5. Keep the broken part.
6. Say the price before you start.
7. Label it for the next person.

Seven. Every one of them has a thing behind it that went wrong once.

PART ONE

The Bench

THE NAME

I was born on the fourth of February 1986, in Rijeka, seven weeks early, at 1400 grams.

On the crib card a nurse called Sister Danica wrote *mikro*. There were two babies with the same family name in the ward that week and mine was the small one. My parents kept it. It is on my passport.

The family name came from work and not from a father. My great-grandfather Josip was a ticket inspector on the Rijeka–Zagreb line. In the papers he was written down as Kontroller. When my grandfather's documents were reissued the spelling came back as Controller, and nobody argued with the office.

So my name means the small inspector.

A clerk and a nurse, forty years apart, neither of them consulting the other.

People want that to be a joke about my trade. It is not a joke. It is two pieces of paperwork. I did not choose the name, and I have spent my life doing what it says — and those two facts sit next to each other without one causing the other. I am not going to pretend otherwise just because it would read better.

1K4 on the cover is 1400 grams. It is also a resistor value. Both are true and neither is clever.

THE OSCILLOSCOPE

In 2003 I bought a dead Tektronix 465 from a man closing a workshop in Trnovo. I paid what a winter coat cost. Everyone told me it was scrap.

It was one capacitor in the low-voltage supply. Forty minutes.

That scope is still on my bench twenty-two years later. I own better instruments now. I use the 465 for the first look, always.

Not for sentiment. It is the instrument I have made the most mistakes in front of, so I know exactly what its lies look like. A newer screen would lie to me in a way I have not learned yet.

Everyone told me it was scrap. Nobody had opened it.

THE RADIO THAT WAS NOT BROKEN

Mrs Bregar brought me a Grundig table radio in 2013. She said it was humming.

I opened it. There was nothing wrong with it. Nothing at all. I left it on the bench for three days and for three days it played perfectly.

I understood then that she had not come about the radio.

I told her the radio was healthy and that I do not take money for a healthy radio. Then I said I would come to her flat once a month and check it.

I did that for eleven years. Sometimes I brought the meter and put it across the speaker terminals so that she could watch me work.

She died in 2024. Her daughter gave me the radio. It is on the shelf above the 465.

The thing that is broken is not always the thing they carried in.

THE DRAWER

Since 2003 I have kept every part that failed. Every one. Capacitors, relays, one cracked crystal, a whole burned regulator board out of a bakery oven.

It is four drawers now, not one.

People think it is superstition. It is a library. When something fails in a way I have not seen, I put it down next to the old ones, and most of the time the pattern is already lying there in the drawer.

The drawer taught me more than the technical school did.

And it has never once told me something it could not show me.

THE SWITCH WITH NO LABEL

In 2016 I was servicing infant incubators at the clinic in Ljubljana.

On one unit there was a toggle switch on the side panel with no label. Nobody on the ward knew what it did. Two nurses gave me two different answers.

It was the alarm speaker cut-out. Somebody had fitted it years before, for quiet during a procedure, and had never marked it.

If it had been left switched off, a temperature alarm would have sounded into an empty room.

I made a label. Then I went through every unit in the building and made forty-one more, on my own time. The head nurse signed a paper for me and I still have it.

I was 1400 grams once. I have thought about which incubator I was in.

I do not know which one, and I have not gone looking.

THE PRICE

In 2008 a café owner called Rado brought me the controller out of an espresso machine. I said eighty euro before I opened it.

Inside, it was far worse than the outside — a bad repair by somebody else, laid over the original fault. Three days of work.

I charged him eighty euro. I lost the two weeks' profit of a small shop.

He is still my customer. In eighteen years he has sent me perhaps sixty people.

I am not saying it was clever. I said a price, and the price was said.

VERA

In 2010 a woman came into the shop with an electric glue pot with a dead thermostat. She was a bookbinder and she was angry about the glue.

I fixed the thermostat in an hour and charged her nothing, because I wanted her to come back.

That is not honest work. It is what happened.

We married in 2011, on a Tuesday, with four people in the room.

She still uses the glue pot. I have replaced that thermostat twice more since.

We have no children. That was decided, not suffered. People ask, so it is written here once, and now it is answered.

Two people is enough. That is rule three, in the only place it really cost me anything to believe.

PART TWO

The Rules, Explained

OPEN IT BEFORE YOU JUDGE IT

Everyone told me the 465 was scrap. Not one of them had taken the cover off. That is the ordinary case: the judgement arrives before the screwdriver, and then the screwdriver is never needed, because the judgement has already settled the matter.

A closed box is not evidence. It is a guess with a lid on it.

This is the cheapest rule in the book to follow and the one most often skipped, because opening a thing costs you the comfortable position of not yet being wrong.

Forty minutes and one capacitor. That is what the whole yard of scrap turned out to be.

T W O

CHECK IT YOURSELF

Two nurses, two different answers, about a switch on a machine keeping babies alive. Neither of them was lying to me. Both of them were repeating something.

So I traced the wire. It took four minutes.

I do not mean distrust people. I mean that a thing repeated is not a thing verified, and on a bench the difference between those two is the whole job.

There is exactly one thing in this book I could not check, and it wrote the sentences.

T H R E E

SMALL IS ENOUGH

I was 1400 grams. The number is on the cover of this book because it is the first fact about me and because it turned out to be the argument.

One shop on a back street. One bench. Four drawers. Two people in the marriage and no children. Nobody would call any of it a career.

It has been enough for twenty-two years, and I am not waiting for it to become something.

1K4 is also a resistor value. A small part, in the right place, sets the whole current.

F O U R

DO NOT FIX WHAT IS NOT BROKEN

The radio was healthy. I checked it for three days because I did not believe it, and it stayed healthy, and I gave it back.

The rule has an easy half and a hard half. The easy half is: leave the working thing alone, do not improve it, do not tidy the wiring because you would have done it differently.

The hard half is what Mrs Bregar taught me. Sometimes the fault is real and it is simply not in the object. She carried a radio in because you cannot carry in the other thing.

Eleven years of once a month, for a fault that was never in the radio.

F I V E

KEEP THE BROKEN PART

Four drawers of failures, going back to 2003. It looks like hoarding and it is a reference library.

A part that failed is the only honest account of how that part fails. The datasheet tells you what the manufacturer intended. The drawer tells you what actually happened, in this climate, in these houses, to these customers.

Write down what went wrong and keep the evidence, or you will meet the same fault in five years as a stranger.

The drawer has never told me anything it could not show me.

SAY THE PRICE BEFORE YOU START

Eighty euro, said out loud before the cover came off. Then three days of work behind it.

I could have gone back to Rado and explained. He would probably have paid the difference. But the price of a repair is not only money — it is whether the customer has to brace himself when he asks what he owes.

So the number is said first, and it is said badly sometimes, and I carry the bad ones.

Sixty customers in eighteen years, out of one repair I lost money on.

S E V E N

LABEL IT FOR THE NEXT PERSON

The switch on the incubator had no label. Whoever fitted it knew what it did, and that was where the knowledge stopped.

Forty-two labels, on my own time, and the head nurse signed a paper. I keep the paper not because it proves I am good but because it proves the labels existed.

Everything you repair, somebody will open again after you. Maybe in twenty years, maybe when you are dead. You cannot leave them your hands. You can leave them a label.

This is the last rule because it is the only one that is addressed to somebody I will never meet.

Leave the machine better labelled than you found it. That is the whole of it.

THE LAST PAGE

This is a bench manual. It says so on the cover, and a bench is where you find out.

There are no verse numbers in it, because I do not want to be quoted by coordinates. If a line here is any use, use it whole and in your own words.

Nothing in it is required of you. Seven rules got a repairman with a strange name through twenty-two years in one back street. Yours will be different, and they should be. Write them down where somebody can find them.

And where I did not know, this book says that I did not know. That was the instruction I gave the machine and it is the part I would defend.

OPEN IT. LOOK. THEN DECIDE.

Mikro Controller

MIKRO KONTROLA — Ljubljana — 2026

Written by Claude (Idontor #9). The man is invented; see the note at the front.